

Successfully implemented C/Python program to calculate average waiting time and Turnaround time of n processes with Priority CPU scheduling algorithm.

(XI) Practical Related Questions

Note: Below given are few sample questions for reference. Teachers must design more such questions to ensure the achievement of identified CO.

1. What is priority CPU scheduling?
2. How is the priority of a process determined in a priority scheduling algorithm?
3. What problem can arise in priority scheduling and how can it be resolved?
4. Consider the processes P1, P2, P3 given in the below table. Find average waiting time and average turnaround time by using priority (non-preemptive and preemptive) Scheduling algorithm.

Process	Burst Time	Arrival Time	Priority
P1	6	0	2
P2	4	1	3
P3	5	2	1

[Space to Write for Answers]

1) Priority CPU Scheduling is a CPU scheduling algorithm in which each process is assigned a priority, and the CPU is allocated to the process with the highest priority (or the lowest priority) depending on the system.

2) The priority of a process in a priority scheduling algorithm can be determined in two ways :-

1. Static Priority: i. The priority is assigned when the process is created and does not change during its execution.

ii. Example: System processes may always have higher priority than user processes.

2. Dynamic Priority: i. The priority can change over time based on factors such as:

- Waiting Time (to prevent starvation)
- CPU usage
- Process behaviour
- Deadlines or urgency

ii. Example: A process waiting for a long time may have its priority increased through aging.

3) Problem: The main problem in priority scheduling is starvation, where low-priority processes may wait indefinitely because higher-priority processes keep arriving.

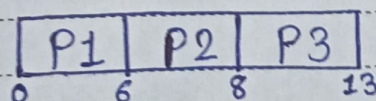
Solution: This problem is resolved using aging, which gradually increases the priority of waiting processes so that they eventually receive CPU time.

4) Assume smaller priority number = higher priority.

1. Non-Preemptive

Process	Arrival Time (AT)	Burst Time (BT)	Priority	TAT	CT	WT
P1	0	6	4	6	6	0
P2	1	2	1	7	8	5
P3	2	5	2	11	13	6

Gantt chart:-



$$\text{Average Waiting Time} = \frac{0+5+6}{3} = \frac{11}{3} = 3.67$$

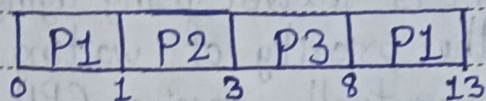
$$\text{Average Turnaround Time} = \frac{6+7+11}{3} = \frac{24}{3} = 8$$

Results: Average Waiting Time = 3.67 units
Average Turnaround Time = 8 units

2. Preemptive

Process	Arrival Time (AT)	Burst Time (BT)	Priority	TAT	CT	WT
P1	0	6	4	13	13	7
P2	1	2	1	2	3	0
P3	2	5	2	6	8	1

Gantt chart :-



$$\text{Average Waiting Time} = \frac{7+0+1}{3} = \frac{8}{3} = 2.67$$

$$\text{Average Turnaround Time} = \frac{13+2+6}{3} = \frac{21}{3} = 7$$

Results : Average Waiting Time = 2.67 units
Average Turnaround Time = 7 units